



Design of a Pupil Tracking System for Robotically Aligned Optical Coherence Tomography

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Background

- Optical coherence tomography (OCT)
 - Non-invasive imaging technology
 - Commonly used to diagnose eye health
- Prone to motion artifacts due to eye or head movement
- Mechanical immobilization limits accessibility
 - Cannot compensate for involuntary eye movement (nystagmus)
- Robotic OCT automatically corrects for motion artifacts in real-time

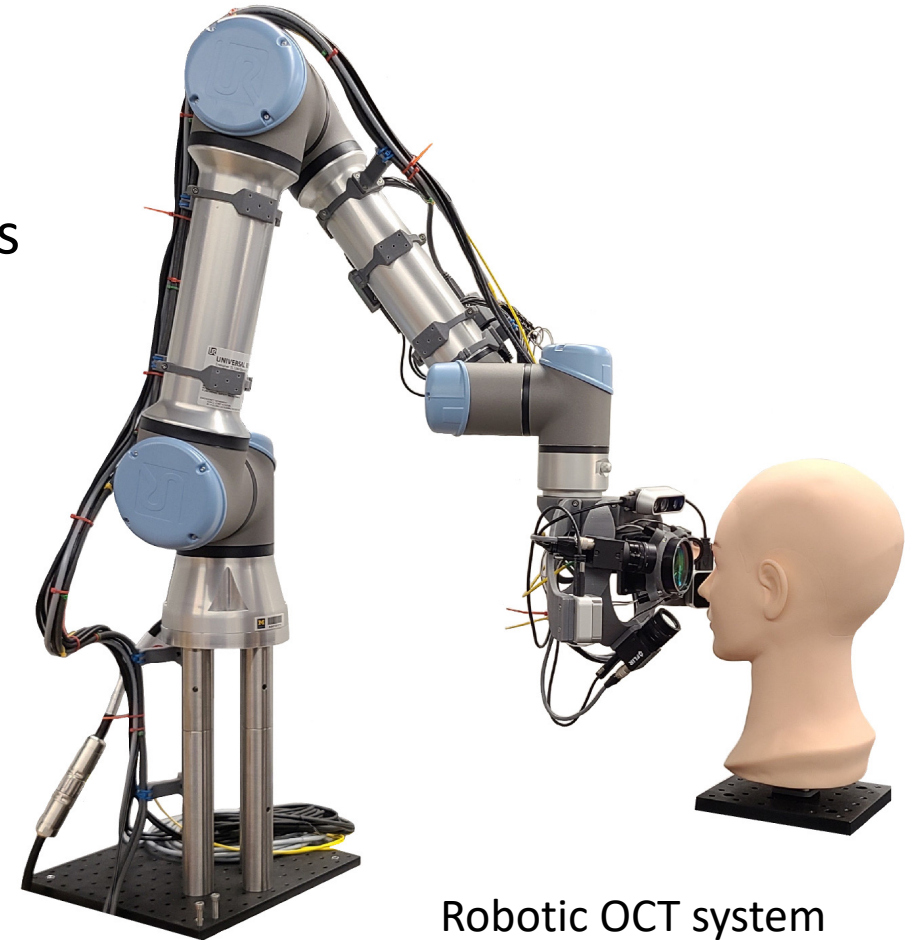


Tabletop OCT system

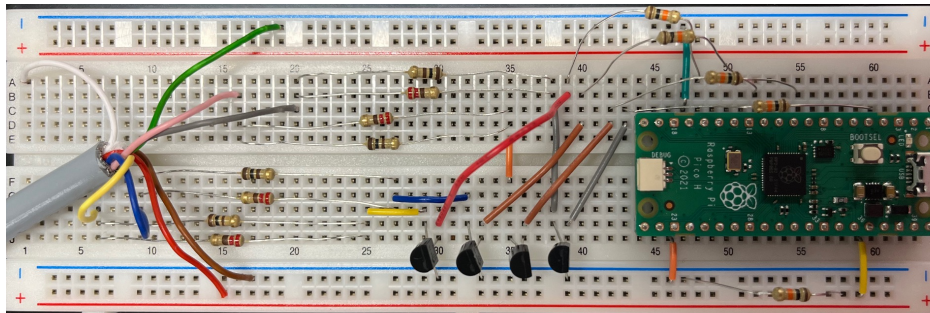
[1] http://link.springer.com/10.1007/978-0-387-68987-6_10

Background

- Pupil tracking maintains alignment with eye
- Current strategy looks for four LED reflections
 - Inefficient pixel search
 - Source of each reflection is ambiguous
- Pulse one LED at a time and compare successive images
 - Efficiently determine reflection location and source

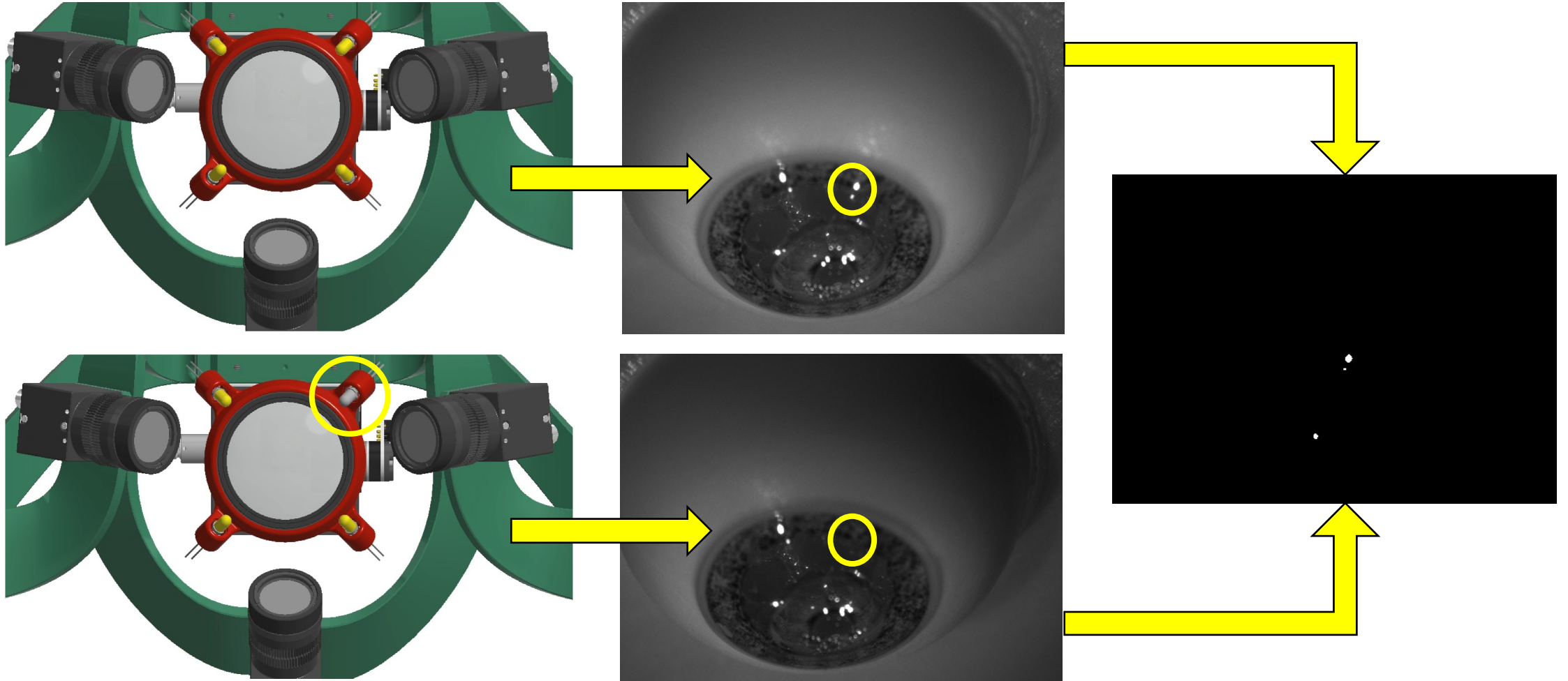


Robotic OCT system



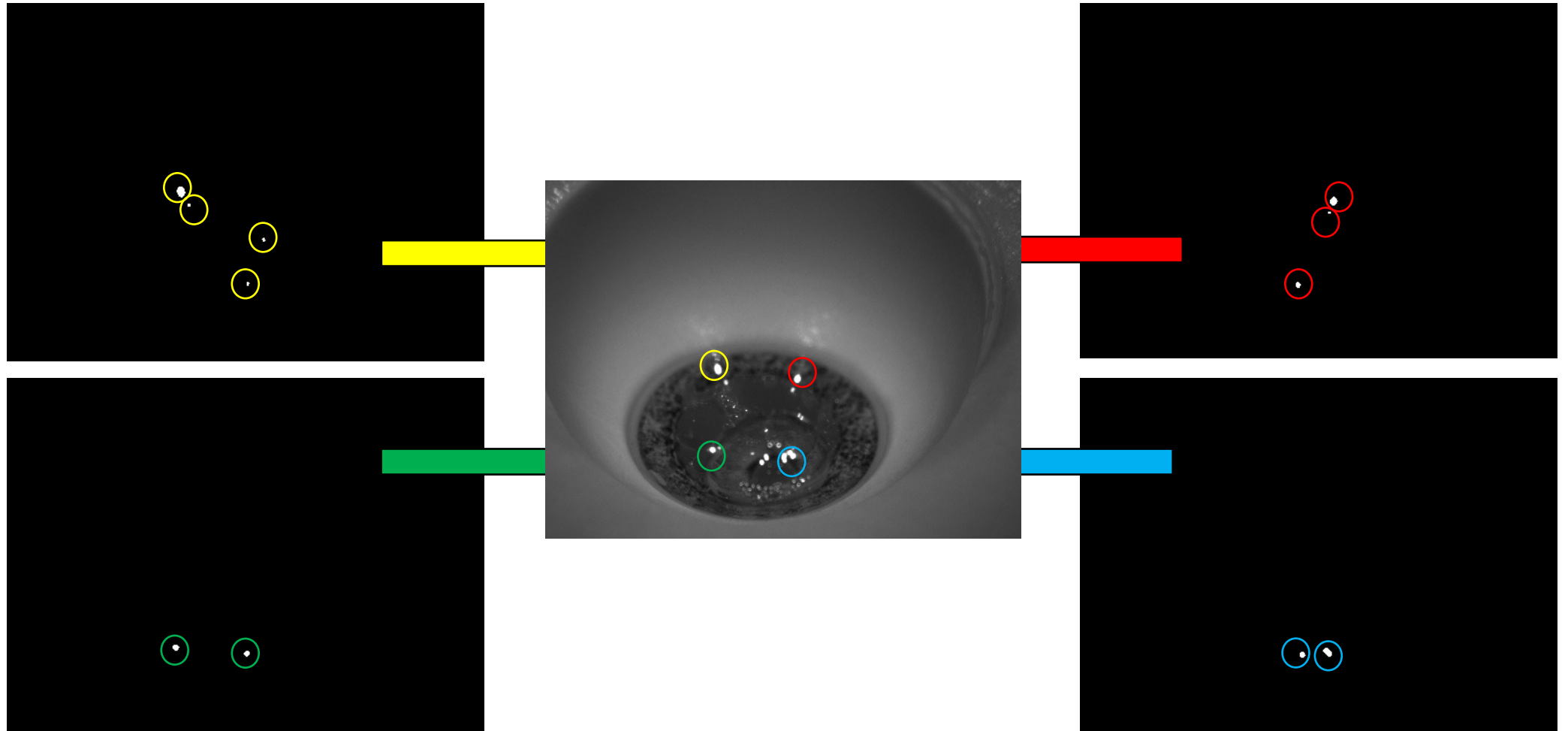
Software Design

Image Processing



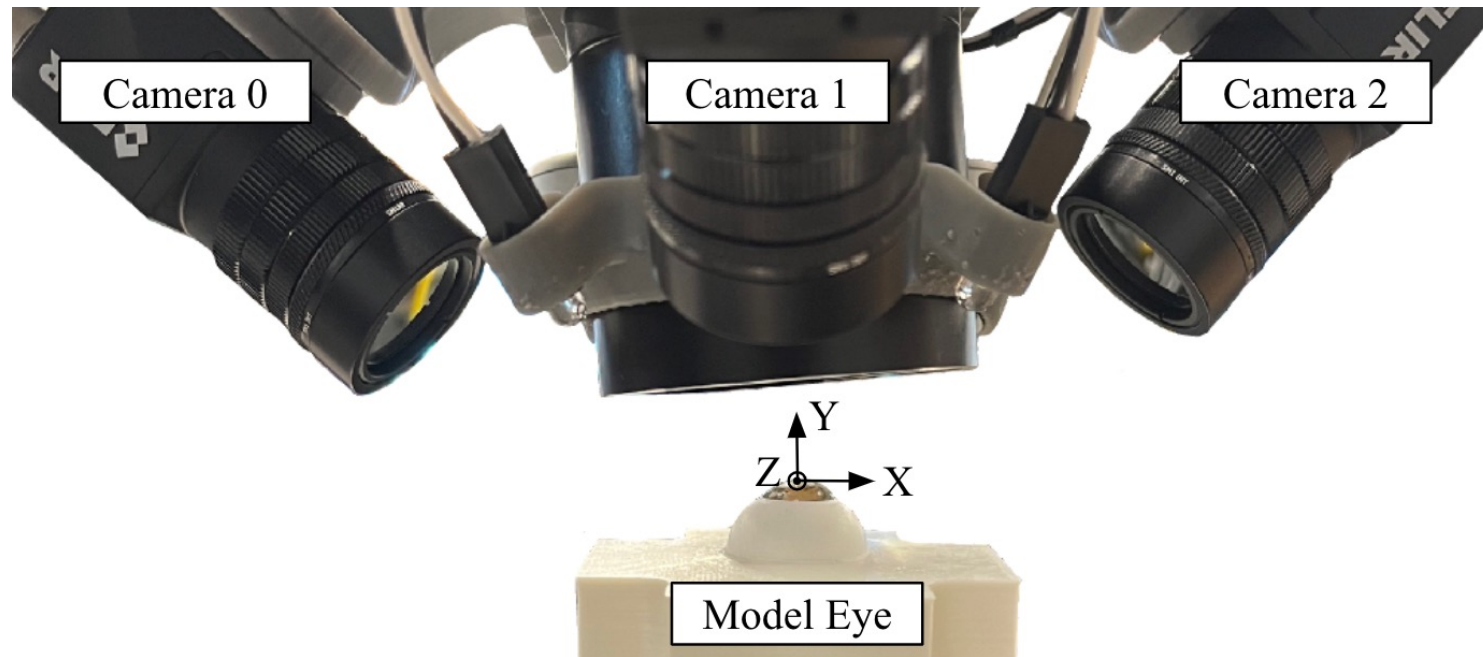
Software Design

Square Heuristic



Experimental Validation

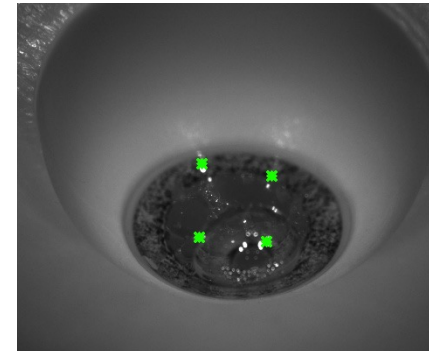
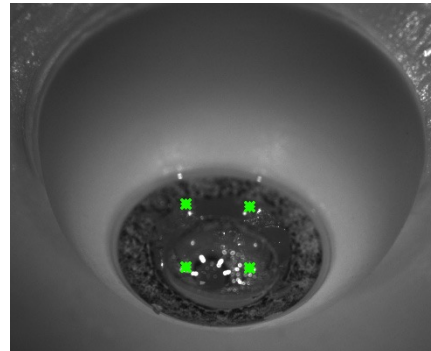
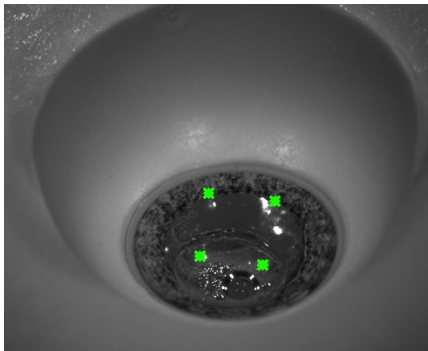
- Test when not optimally aligned to eye
- Rotated end effector $[-15, 15]$ degrees in 5 degree increments around each coordinate axis



Results

- 95% true positive rate across all angles and axes
- Effective beyond the typical amplitude of nystagmus (3–4°)

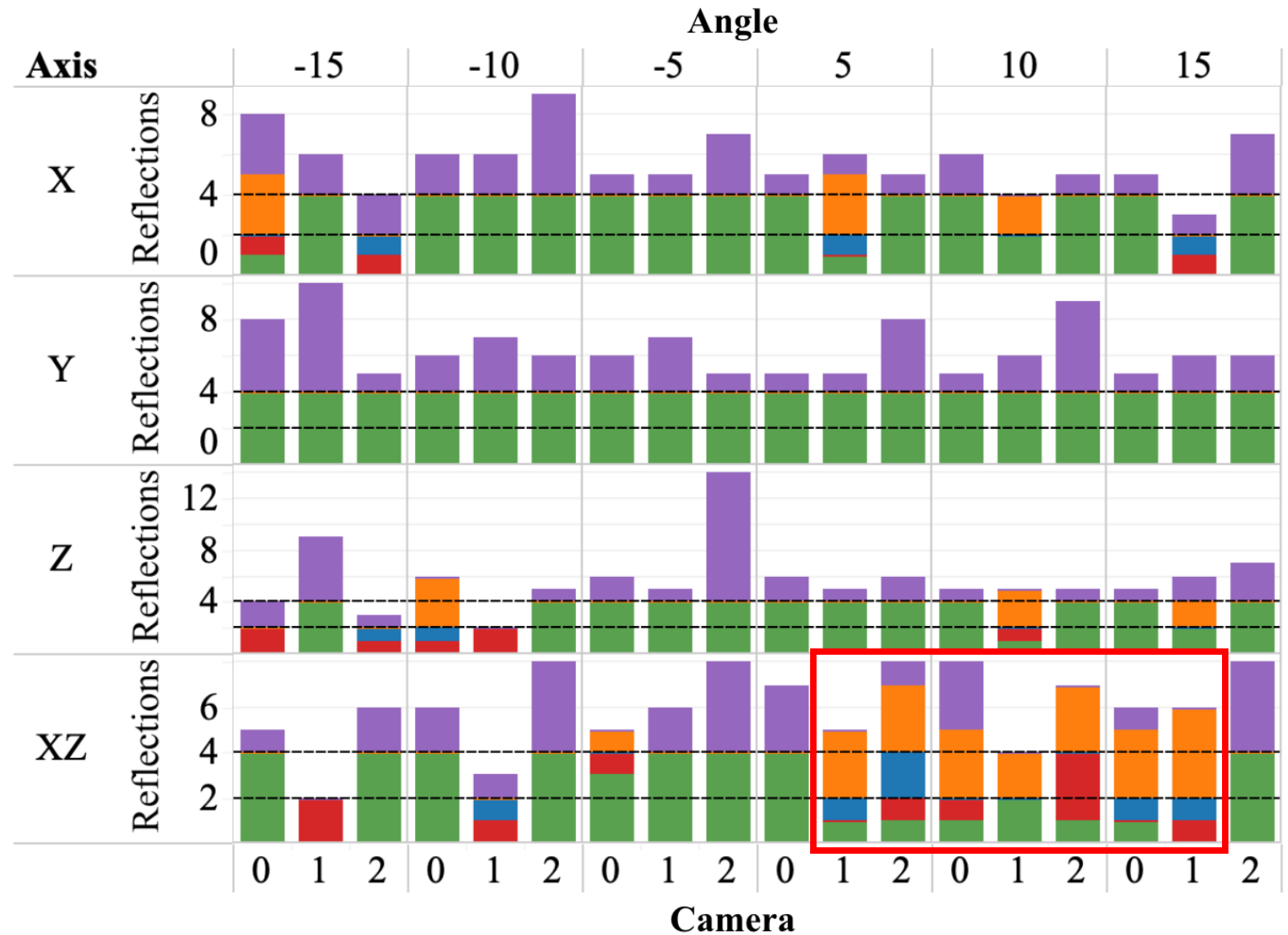
		Actual	
		True	False
Predicted	True	95%	23%
	False	5%	77%



Results

Accuracy of LED Localization With Rotational Offsets

■ True Negative
 ■ False Positive
 ■ False Negative
 ■ True Positive
 ■ Miss

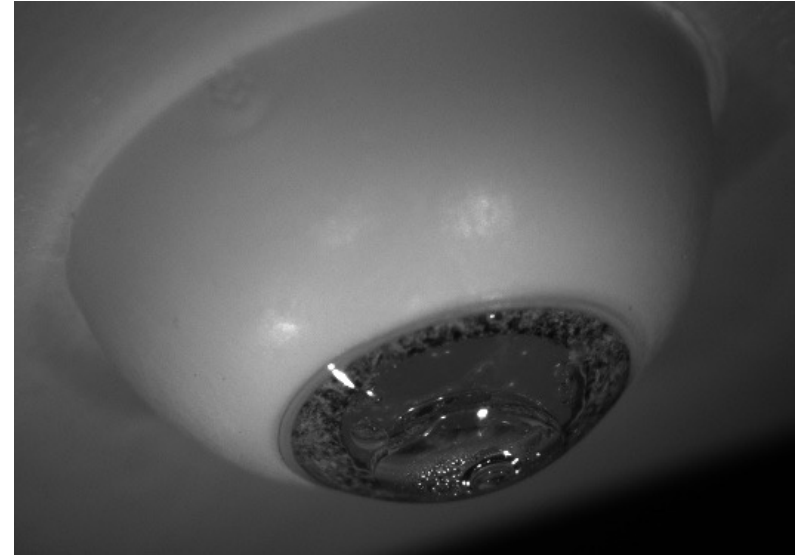


Limitations

- Square heuristic always returns four reflections
- Not all reflections are visible from side view of eye
- Best case: two true positives, two false positives
- Worst case: no reflections found



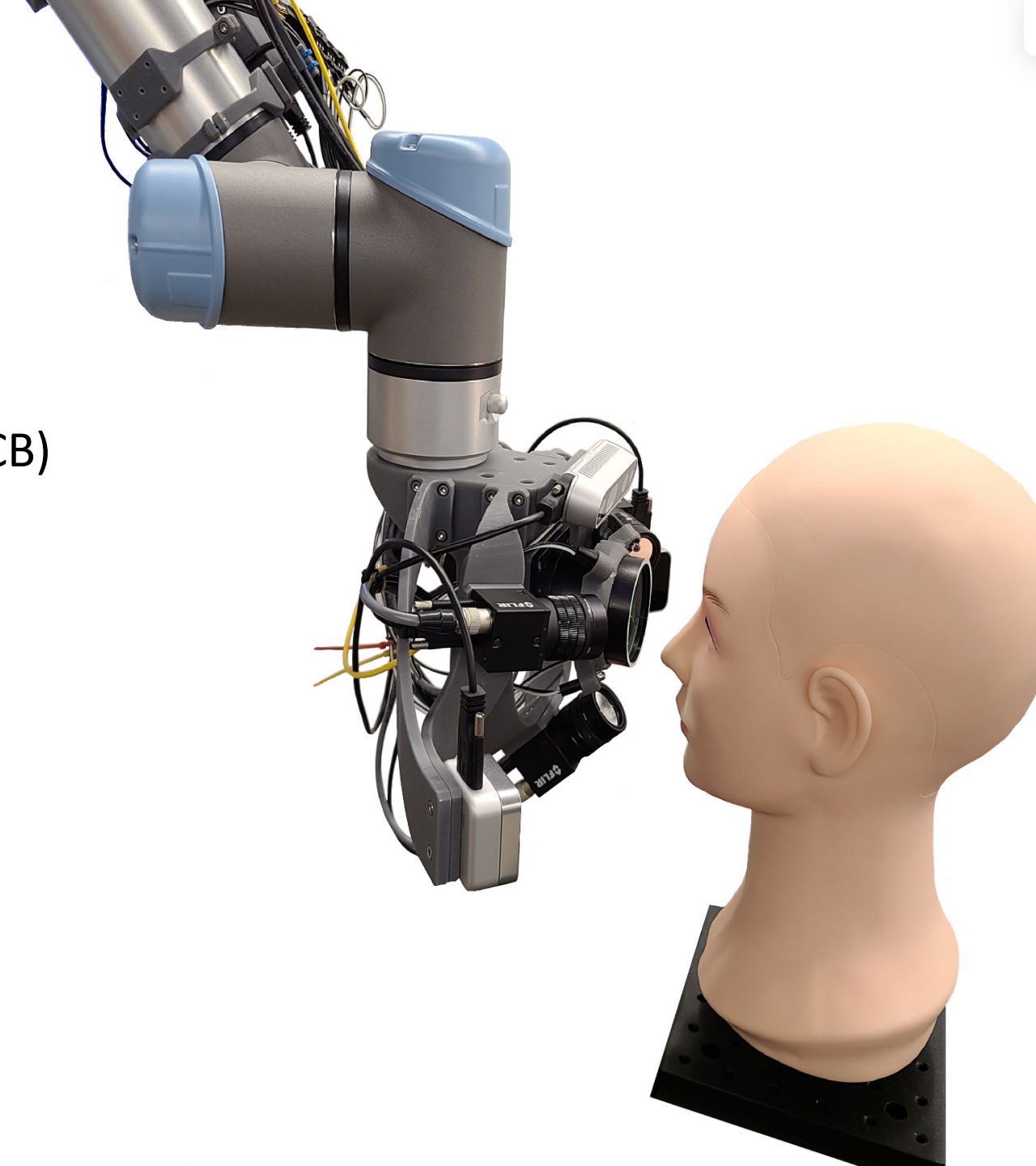
Zero degrees rotation (full view)



15 degrees rotation about X and Z axes (partial view)

Future Work

- Implement pair detection
- Optimize sampling frequency
- Mount trigger circuit on robot (PCB)



Acknowledgements

Thank you to Dr. Mark Draelos and the Image-Guided Medical Robotics Lab for your mentorship and support throughout this project!

